Object Oriented Programming Lab (CSEG1044) Experiment-05&06: Arrays + String Handling (Student Handout — Before Submission)

Tofik Ali

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1 About this handout

This is a **pre-submission** handout. It specifies what to implement and what to submit.

2 Experiment 05 (Syllabus)

- **Objective:** To work with one-dimensional and two-dimensional arrays.
- **Problem Statement:** Develop Java programs to perform operations on arrays.
- **Expected Outcome:** Students will be able to store and process data using arrays.

3 Experiment 06 (Syllabus)

- **Objective:** To perform string manipulation using Java String classes.
- **Problem Statement:** Implement string operations using built-in Java methods.

- **Expected Outcome:** Students will be able to manipulate textual data.

4 What you must do (minimum requirements)

4.1 Task A — Array operations (Exp-05)

1. Create `MarksAnalysis.java`:
   - read `n` and then `n` marks into an `int[]`,
   - print min, max, sum, average,
   - print `passCount` where marks ≥ 40.
2. Create `MatrixAdd.java`:
   - read rows `r` and cols `c`,
   - read matrix `A` and matrix `B` of size `r x c`,
   - print `A + B` (element-wise sum).

4.2 Task B — String toolkit + palindrome (Exp-06)

Create `StringToolkitDemo.java` and demonstrate:

1. `==` vs `equals()` using:
   - `new String("upes")`
   - string literals like `"java"`
2. Immutability:
   - show that calling `toUpperCase()` does not change the original string unless stored.
3. Palindrome checker:
   - read a full line (sentence),
   - ignore spaces and case,
   - print whether it is a palindrome.

5 Hints (read only if stuck)

- Average should be computed using `sum / (double) n` (avoid integer division).
- For 2D arrays: `a.length` is number of rows; `a[i].length` is number of columns in row `i`.
- Use `equals()` for value comparison of strings; `==` checks references.
- Use `StringBuilder` when repeatedly appending in a loop.

6 What to submit (MANDATORY)

Submit a single zipped folder containing:

- `MarksAnalysis.java`
- `MatrixAdd.java`
- `StringToolkitDemo.java`
- sample output for all programs
- observations (5–10 lines): what you learned about arrays and string comparison

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed March 3, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed March 3, 2026).